

Bhavesh Ghanchi

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Education

University of Southern California

Aug 2024 – May 2026

Master of Science, Computer Science (GPA: 3.83/4.0) | Los Angeles, CA

K. J. Somaiya College of Engineering

Aug 2020 – Jun 2024

Bachelor of Technology, Computer Engineering (GPA: 9.64/10) | Mumbai, India

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, C#, SQL

Backend and Systems: Node.js, Express.js, FastAPI, RESTful APIs, PostgreSQL, MongoDB, Firebase, Supabase

Frontend: React, Next.js, Tailwind CSS, HTML/CSS

AI/ML: PyTorch, TensorFlow, Transformers, LLMs, RAG, Fine-tuning (QLoRA), FAISS, NLP, OpenAI API

Infrastructure and Tools: Docker, AWS, GCP, Git, GitHub Actions, Jenkins, Linux, ArcGIS, Unity

Experience

Easley-Dunn Productions | Software Engineer Intern | Torrance, CA (Remote)

Jul 2026 – Present

- Building backend integrations and persistence for *Robot Race* across Firebase, Unity Gaming Services, Mirror, and EOS.
- Developing core gameplay systems in C# on a state-machine architecture, smoothing level transitions and control response.
- Diagnosing player movement and collision defects via code analysis, runtime logging, and map-specific testing to improve stability.

Streets Are For Everyone (SAFE) | Software Developer Intern | Los Angeles, CA (Remote)

Jun 2025 – Aug 2025

- Classified a 170+ route LA Metro bike dataset in ArcGIS by bikeway type to produce safety-rated cycling maps.
- Built a React and Node.js web app serving the classified data, replacing the site's static map.
- Developed a route-recommendation chatbot on the OpenAI API surfacing safer paths from the classified GIS layer.
- Released to a beta cohort of 60 cyclists and incorporated their feedback ahead of wider rollout.

Colgate-Palmolive | Full-Stack Intern | Mumbai, India

Jan 2024 – Jun 2024

- Shipped React, TypeScript, and Node.js features for a consumer platform serving 10,000+ users across 5 regions.
- Optimized RESTful APIs and PostgreSQL queries, reducing API latency by 25% under peak traffic.
- Architected a CDN-backed multi-region deployment with 4-language localization to support international rollout.
- Expanded Jest automated test coverage by 50%, reducing recurring production regressions.

K. J. Somaiya College of Engineering | Research Intern | Mumbai, India

Mar 2023 – May 2023

- Built a multi-agent NLP pipeline in Python automating text extraction and preprocessing across 50+ clinical PDFs.
- Developed a parallel-processing named-entity recognition system achieving 90%+ entity extraction accuracy on biomedical corpora.

SayHey | Backend Developer Intern | Mumbai, India

Jun 2022 – Aug 2022

- Owned backend services for a scheduling platform with 5,000+ users, maintaining 99%+ uptime.
- Designed and optimized REST APIs handling 500+ concurrent sessions, reducing booking latency by 30%.

Projects

OTAS: Observability Tool for Agentic AI Systems (Team project) | Python, FastAPI, LLM APIs, PostgreSQL, Docker | GitHub | Apr 2026

- Built an observability layer capturing agent decisions, prompts, tool calls, and latency across 10+ agents.
- Developed a real-time dashboard tracking 800+ agent tasks per day with anomaly detection on execution traces.

LEAP: Loss-aware End-to-end Acknowledged Protocol | Java, Maven, UDP sockets | GitHub | Blog Series | May 2026

- Engineered a TCP-like reliable transport over UDP in Java with sliding window, fast retransmit, adaptive RTO, and AIMD.
- Achieved 46 MB/s on 10 MiB transfers at 0% loss, with end-to-end SHA-256 integrity verified on every run.
- Sustained 88.8% efficiency at 10% packet loss across a 3-trial sweep with documented abort behavior beyond it.

Biomedical RAG Platform (Team project) | PyTorch, FAISS, QLoRA, Transformers | HuggingFace | Apr 2025

- Built a retrieval-augmented generation pipeline with FAISS vector search over biomedical literature embeddings.
- Fine-tuned a biomedical LLM using QLoRA, improving answer quality by +5.6% BERTScore F1 over the base model.

PrepLoop: Interview Preparation Platform | Next.js, TypeScript, Supabase, PostgreSQL, Vercel | Live Site | GitHub | Jul 2026

- Developed cross-device sync with Supabase Auth, PostgreSQL row-level security, and offline caching.
- Built approach-first coding, system-design, and behavioral workflows, deployed via GitHub Actions and Vercel.